

** So important
chapter*

Module Flow

- IDS vs IPS

- False +ve / False -ve

- Signature vs anomaly detection

- IDS Placement

- HIDS vs NIDS

- Snort

- Active vs Passive IDS

1

**Understand Different
Types of Network
Segmentation**

2

**Understand Different
Types of Firewalls
and their Role**

3

**Understand Different
Types of IDS/IPS and
their Role**

4

**Understand Different
Types of Honeypots**

5

**Understand Different
Types of Proxy Servers
and their Benefits**

6

**Discuss Fundamentals
of VPN and its
importance in Network
Security**

7

**Discuss Security Incident
and Event Management
(SIEM)**

8

**Discuss User Behavior
Analytics (UBA)**

9

**Understand Various
Antivirus/Anti-
malware Software**

Intrusion Detection and Prevention System (IDS/IPS)

Stands for

01

An intrusion detection and prevention system (IDS/IPS) is a network security appliance that **inspects all inbound and outbound network traffic** for **suspicious patterns** that might indicate a network or system security breach

- intrusion
- Attack
- security breach

Untrusted Network



02

If found, the IDS will alert the administrator about the **suspicious activities**

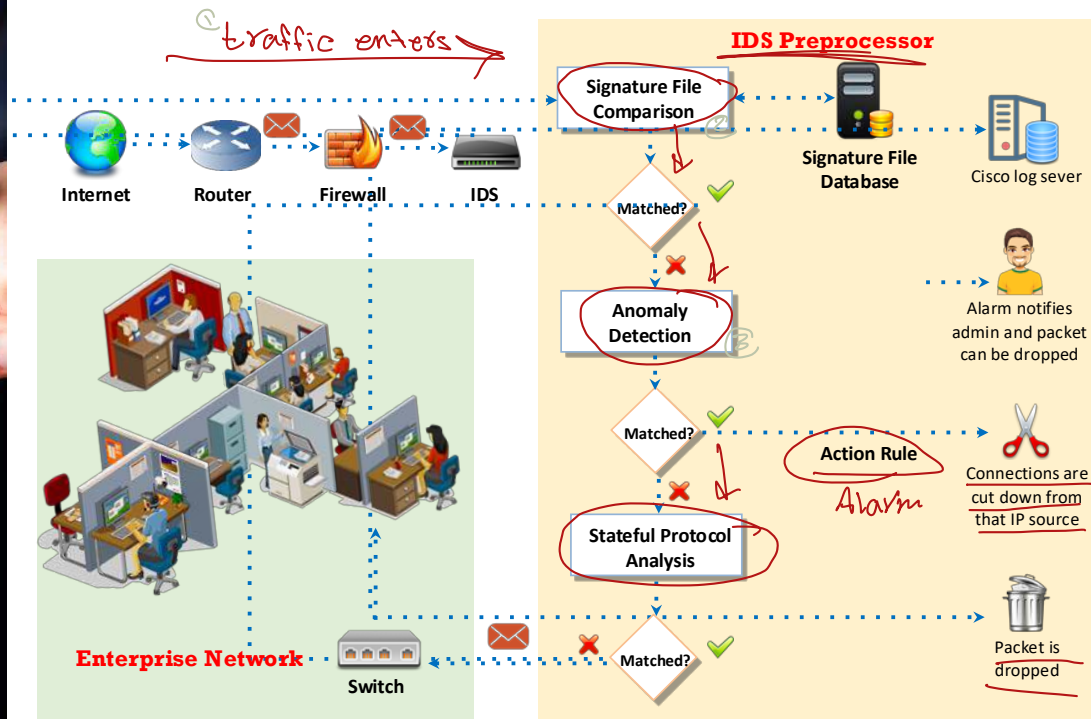
03

IDS checks the network traffic for **signatures** that match known intrusion patterns and triggers an alarm when a match is found

IDS	IPS
Detects	Detects+Block
Send alerts	takes action

IDS

How does an IDS Work?



Role of an IDS in Network Defense



*Firewall Filters traffic
IDS analyze traffic deeply
-IDS helps detect attacks
that bypass firewalls



- ☐ An IDS works from inside the network, unlike a firewall which only looks outside the network for intrusions

- ☐ An IDS is placed behind the firewall, inspecting all the traffic, looking for heuristics and a pattern match for intrusions

How an IDS Detects an Intrusion?

3 methods

Signature Recognition

Signature recognition, also known as misuse detection, tries to **identify events** that indicate an abuse of a system or network resource

look for: Known attack Patterns

ex: if Packet matches SQL injection signature, IDS alerts admin

Protocol Anomaly Detection

In this type of detection, models are built to explore **anomalies** in the way in which vendors deploy the **TCP/IP specification**



Anomaly Detection

It detects the **intrusion based** on the fixed behavioral characteristics of the users and components in a computer system

IDS learns: Normal network behavior, then detects deviations

ex: Sudden huge bandwidth usage

IDS Capabilities



- ❑ IDS provides an additional layer of security to the network under the **defense-in-depth** principle
- ❑ IDS does several things that basic **firewalls** cannot do
- ❑ IDS helps minimize the chance of **missing security threats** that could come from firewall evasions

IDS/IPS Functions

- | | |
|--|---|
| ✓ Monitoring and analyzing both user and system activities | ✓ Recognizing typical attack patterns |
| ✓ Analyzing system configurations and vulnerabilities | ✓ Analyzing abnormal activity patterns |
| ✓ Assessing system and file integrity | ✓ Tracking user policy violations |

IDS/IPS Limitations:

What an IDS/IPS is NOT?

**Network Logging
Systems**

**Vulnerability
Assessment Tools**

**IDS/IPS cannot act
as or replacement of**

**Antivirus
Products**

**Cryptographic
Systems**

IDS/IPS Security Concerns



- ❑ Improper IDS/IPS configuration and management will make an IDS/IPS **ineffective**
- ❑ IDS/IPS deployment should be done with careful planning, preparation, prototyping, testing, and specialized training

Common Mistakes in IDS/IPS Configuration

- ✓ Deploying an IDS in a location where it **does not see** all the network traffic
- ✓ Frequently **ignoring the alerts** generated by the IDS
- ✓ **Not having** the proper **response policy** and the best possible solutions to deal with an event
- ✓ Not fine-tuning the IDS for **false negatives** and **false positives**
- ✓ Not updating the IDS with the **latest new signatures** from the vendor
- ✓ Only monitoring **inbound connections**



General Indications of Intrusions

File System Intrusions

- ☐ The presence of new or **unfamiliar files**, or programs
- ☐ Changes in **file permissions**
- ☐ Unexplained changes in a file's **size**
- ☐ **Rogue files** on the system that do not correspond to the master list of signed files
- ☐ **Missing files**



Network Intrusions

- ☐ **Repeated probes** of the available services on your machines
- ☐ Connections from **unusual locations**
- ☐ Repeated login attempts from **remote hosts**
- ☐ A sudden **influx of log data**

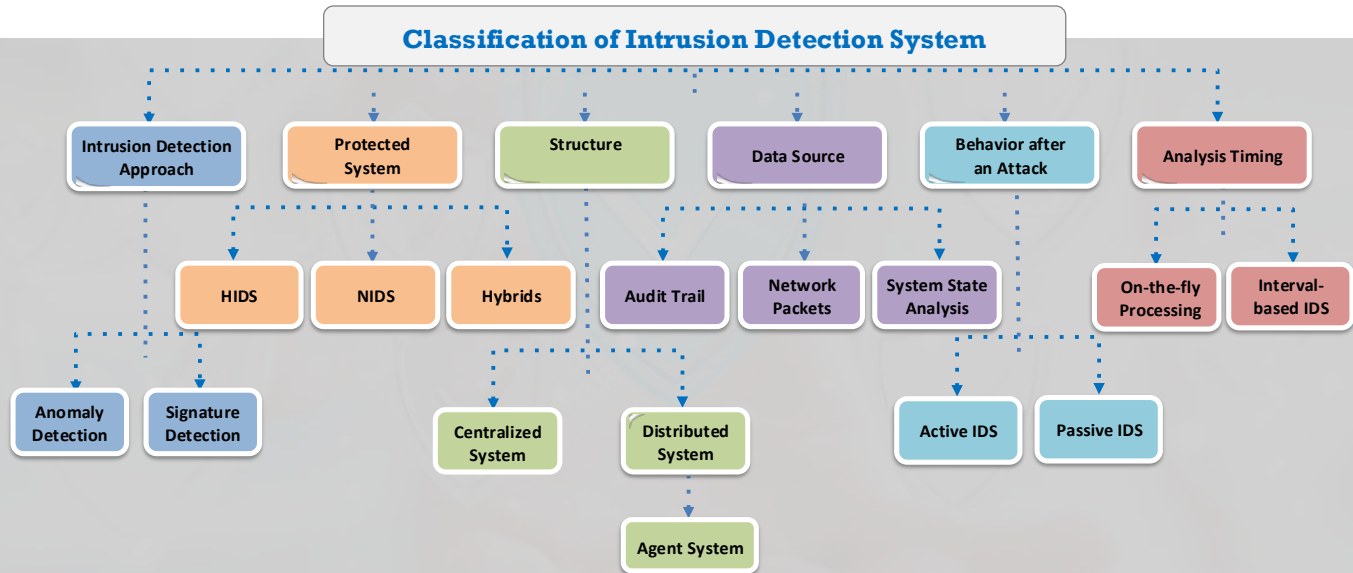


System Intrusions

- ☐ **Short** or incomplete logs
- ☐ Unusually **slow** system performance
- ☐ **Missing** logs or logs with incorrect permissions or ownership
- ☐ **Modifications** to system software and configuration files
- ☐ Unusual **graphic displays** or text messages
- ☐ **Gaps** in system accounting
- ☐ System crashes or **reboots**
- ☐ **Unfamiliar** processes

IDS Classification

An IDS is **classified** based on an approach, protected system, structure, data source, behavior, and time analysis



Approach-based IDS

Signature-Based Detection

- ❑ Known as **misuse detection**
- ❑ **Monitors** patterns of data packets in the network and compares them to pre-configured network attack patterns, known as signatures
- ❑ This method uses string comparison operations to compare **ongoing activity**, such as a packet or a log entry, against a list of signatures

Advantages

- It detects attacks with minimal false alarms
- It can **quickly** identify the use of a specific tool or technique
- It assists administrators to quickly track any potential **security issues** and initiate incident handling procedures

Disadvantages

- This approach only detects **known threats**, the database must be updated with new attack signatures constantly
- It utilizes tightly defined signatures that prevent it from detecting **common variants** of the attacks

Approach-based IDS (Cont'd)

Anomaly-based Detection

- ❑ In this approach, alarms for **anomalous** activities are generated by evaluating network patterns such as what sort of bandwidth is used, what protocols are used, and what ports and which devices are connected to each other
- ❑ An IDS monitors the typical activity for a particular time interval and then builds the **statistics** for the network traffic
- ❑ For example: anomaly-based IDS monitors activities for normal Internet bandwidth usage, failed logon attempts, processor utilization levels, etc.



Advantages

- ✓ An anomaly-based IDS identifies **abnormal** behavior in the network and detects the symptoms for attacks without any clear details
- ✓ Information acquired by anomaly detectors is further used to define the signatures for **misuse detectors**



Disadvantages

- ✓ The rate of generating **false alarms** is high due to unpredictable behavior of users and networks
- ✓ The need to create an **extensive set of system events** in order to characterize normal behavior patterns

Approach-based IDS (Cont'd)



This method **compares** observed events with predetermined profiles based on accepted definitions of benign activity for each protocol to identify any deviations of the protocol state



It also detects variations in command length, minimum/maximum values for **attributes** and other **potential anomalies**

Stateful Protocol Analysis



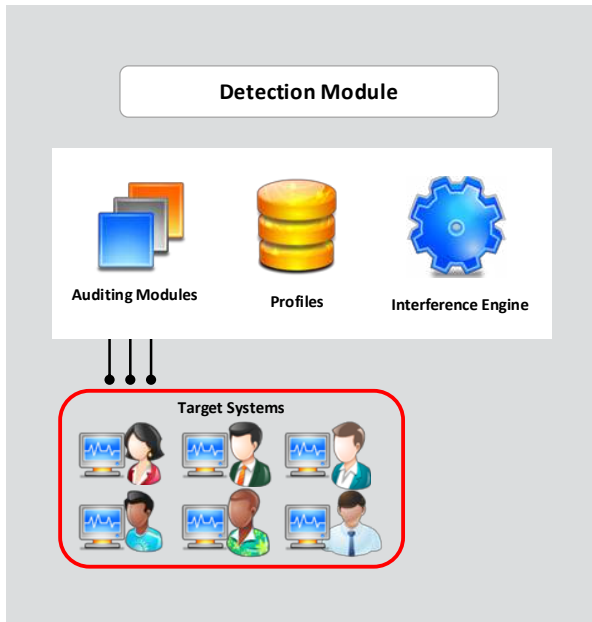
It can identify unpredictable sequences of commands. For example, it can identify activities such as issuing the same commands repeatedly or arbitrary commands being used



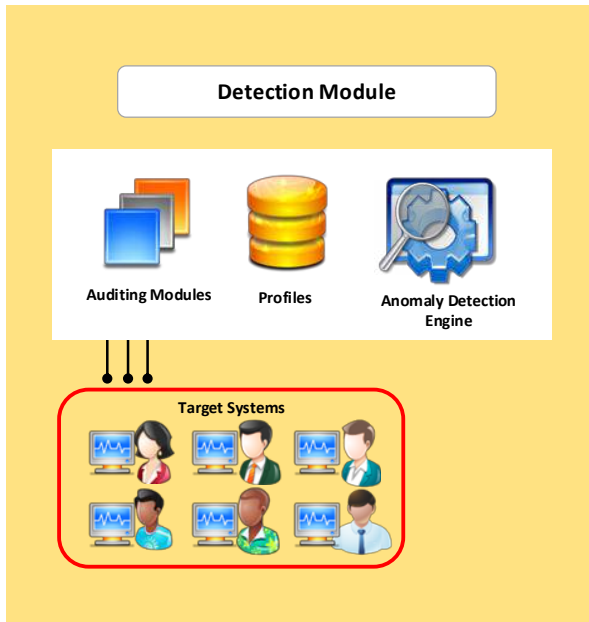
For any protocol performing **authentication**, the IDS/IPS will keep track of the authenticator being used for each session and will record the authenticator involved in the suspicious activity

Anomaly and Misuse Detection Systems

Misuse Detection System

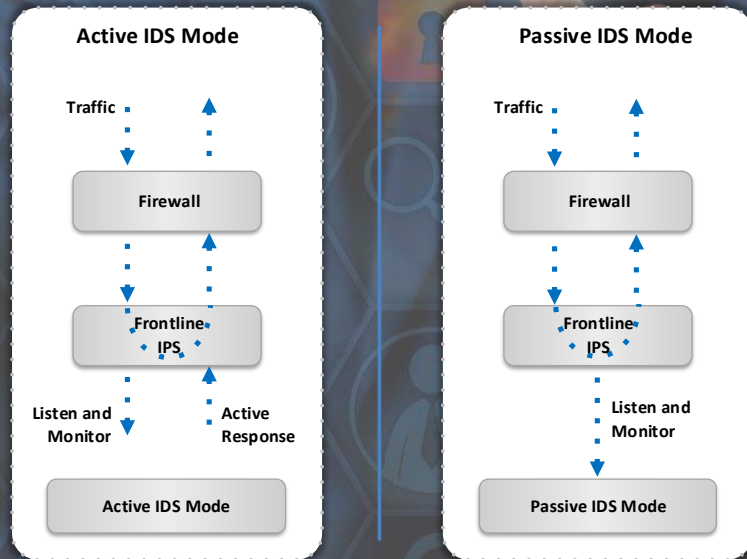


Anomaly Detection System



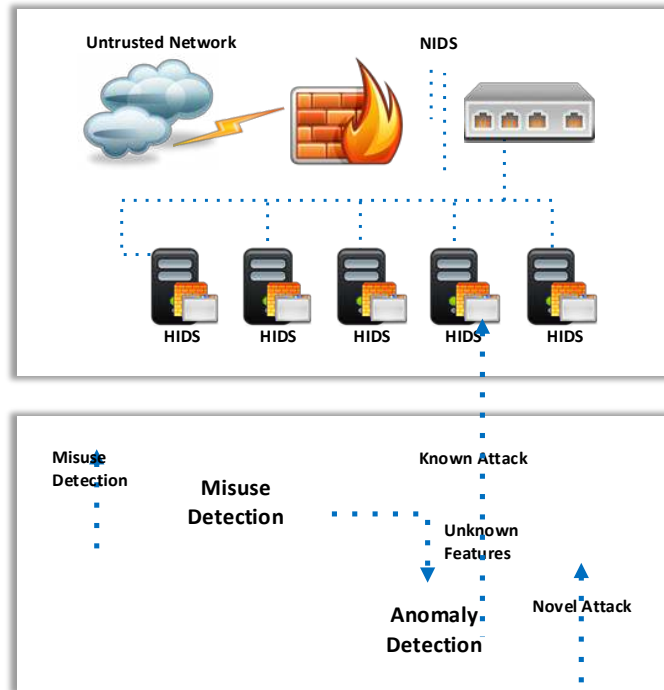
Behavior-based IDS

- ❑ An IDS is categorized based on how it reacts to a **potential intrusion**
- ❑ It functions in one of two modes, active or passive, based on the behavior after an attack
- ✓ **Active IDS: Detects and responds** to detected intrusions
- ✓ **Passive IDS: Only detects** intrusions



Protection-based IDS

- An IDS is classified based on the system/network it offers **protection** to
 - If it protects the network, it is called a network intrusion detection system (**NIDS**)
 - If it protects a host, it is called a host intrusion detection system (**HIDS**)
 - If it protects the network and a host, it is called a hybrid intrusion detection system (**Hybrid IDS**)
- A hybrid IDS combines the advantages of both the **low false-positive rate** of a NIDS and the anomaly-based detection of a HIDS to detect unknown attacks



Structure-based IDS



An IDS is also classified as a **centralized IDS** or a **distributed IDS**, this classification is based on the structure of the IDS



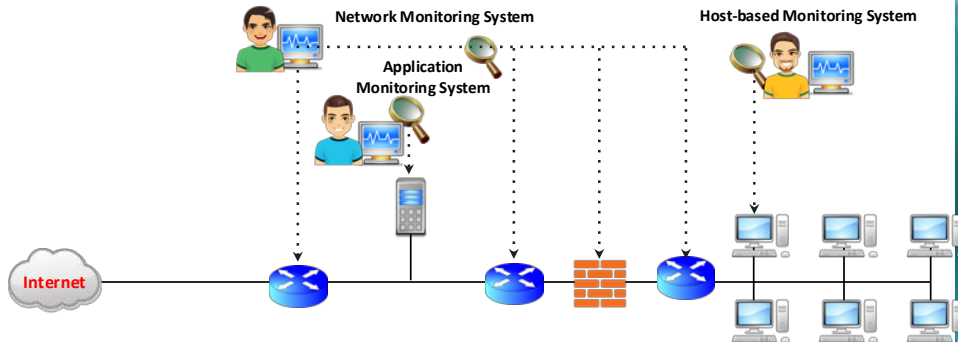
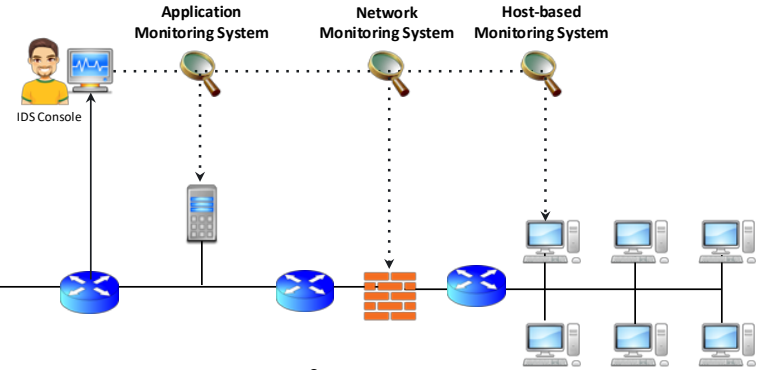
In a centralized IDS, all data is shipped to a **central location** for analysis, independent of the number of hosts that are monitored



In a distributed IDS, **several IDS** are deployed over a large network and each IDS communicates with each other for traffic analysis

Structure-based IDS (Cont'd)

Centralized Control



Fully Distributed (Agent-based) Control



Analysis Timing-based IDS



Analysis time is a **span of time elapsed** between the events occurring and the analysis of those events



An IDS is Categorized based on Analysis Time as:



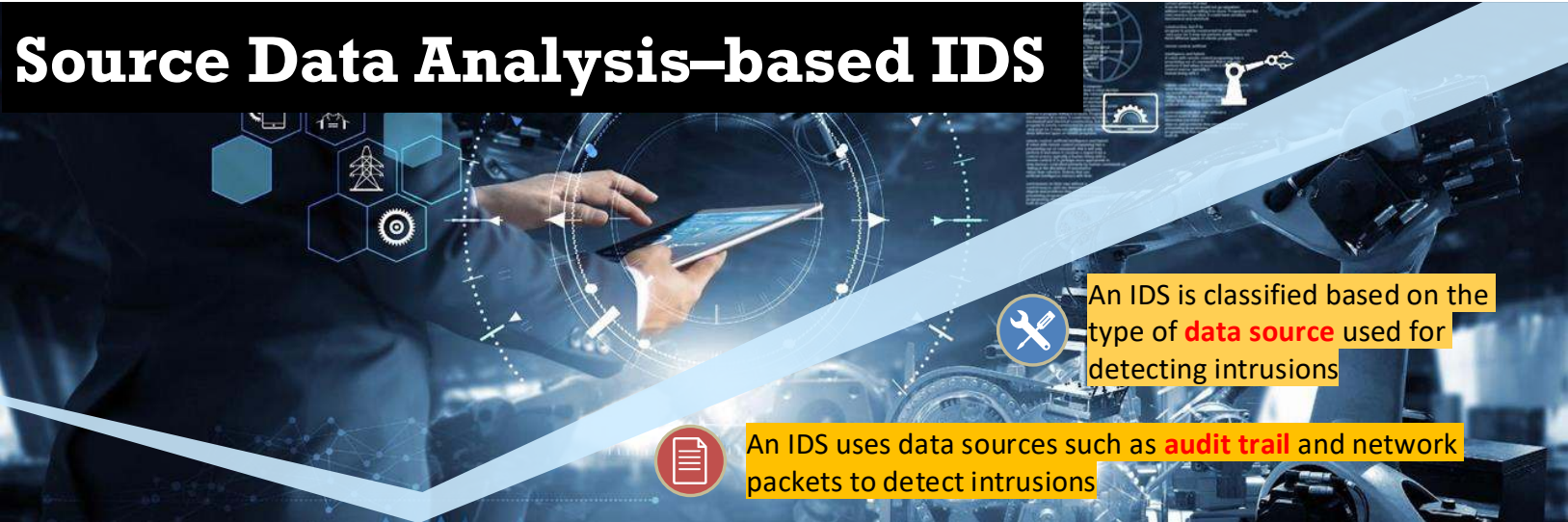
Interval-based IDS

- ☐ The information about an intrusion detection does not flow continuously from monitoring points to analysis engines, it is simply **stored and forwarded**
- ☐ It performs analysis of the detected intrusion **offline**

Real-Time-based IDS

- ☐ The information about an intrusion detection **flows continuously** from monitoring points to analysis engines
- ☐ It performs analysis of the detected intrusion **on the fly**

Source Data Analysis–based IDS



An IDS is classified based on the type of **data source** used for detecting intrusions



An IDS uses data sources such as **audit trail** and network packets to detect intrusions

Intrusion Detection Using Audit Trails

- ☐ **Audit trails** help the IDS detect performance problems, security violations, and flaws in applications



Intrusion Detection Using Network Packets

- ☐ Capturing and analyzing **network packets** help an IDS detect well-known attacks



IDS Components



An IDS system is built on various components. Knowledge of their **functions** and **placement** is required for effective IDS implementation

IDS Components



Network Sensors

Alert Systems

Command Console

Response System

Attack Signature Database



Network Sensors



Network sensors are hardware and software components that **monitor** network traffic and trigger **alarms** if any abnormal activity is detected

Network sensors should be placed and located at common entry points in a network such as:

- ☐ Internet gateways
- ☐ In between LAN connections
- ☐ Remote access servers used to receive dial-up connections
- ☐ VPN devices
- ☐ Either side of firewall

The image displays two screenshots of the SGUIL-0.9.0 interface, which is a network security monitoring tool. The top screenshot shows the 'RealTime Events' tab, displaying a table of alerts. The bottom screenshot shows the 'Escalated Events' tab, displaying a table of alerts with more details.

RealTime Events Table:

ST	CNT	Sensor	Alert ID	Date/Time
RT	1	bob-virtua...	3.3273	2019-09-25 02:17:06
RT	2	bob-virtua...	3.3292	2019-09-25 03:23:07
RT	2	bob-virtua...	3.3293	2019-09-25 03:23:07
RT	1	bob-virtua...	3.3294	2019-09-25 03:23:12
RT	24	bob-virtua...	3.3301	2019-09-25 03:25:30
RT	24	bob-virtua...	3.3302	2019-09-25 03:25:30
RT	1	bob-virtua...	3.3319	2019-09-25 03:25:31
RT	0	bob-virtua...	3.3342	2019-09-25 03:25:31
RT	12	bob-virtua...	3.3354	2019-09-25 06:36:37
RT	12	bob-virtua...	3.3366	2019-09-25 06:36:37
RT	12	bob-virtua...	3.3378	2019-09-25 06:36:37
RT	23	bob-virtua...	3.3390	2019-09-25 09:50:11
RT	9	bob-virtua...	3.3392	2019-09-25 09:52:05
RT	36	bob-virtua...	3.3434	2019-09-25 11:06:32
RT	20	bob-virtua...	3.3436	2019-09-25 11:06:32

Escalated Events Table:

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	3	bob-virtua...	1.1752	2019-09-19 01:14:51	0.0.0.0		10.10.10.16		0	[OSSEC] Windows: System time changed.
RT	30	bob-virtua...	1.1402	2019-09-09 08:23:05	0.0.0.0		10.10.10.16		0	[OSSEC] Windows: Login Failure - Unknown us...
RT	2	bob-virtua...	1.1749	2019-09-19 01:09:34	0.0.0.0		10.10.10.16		0	[OSSEC] The audit log was cleared
RT	2	bob-virtua...	1.1751	2019-09-19 01:42:16	0.0.0.0		10.10.10.16		0	[OSSEC] PAM: User login failed.
RT	2	bob-virtua...	1.1748	2019-09-19 01:08:48	0.0.0.0		10.10.10.16		0	[OSSEC] Listened ports status (netstat) change...
RT	60	bob-virtua...	1.1853	2019-09-19 01:00:00	0.0.0.0		10.10.10.16		0	[OSSEC] Integrity checksum changed.
RT	4	bob-virtua...	1.1849	2019-09-19 00:58:55	0.0.0.0		10.10.10.16		0	[OSSEC] Host-based anomaly detection event (...)
RT	35	bob-virtua...	1.1857	2019-09-19 01:00:03	0.0.0.0		10.10.10.16		0	[OSSEC] File added to the system.
RT	1	bob-virtua...	3.3995	2019-12-24 12:11:06	10.10.10.50	61121	10.10.10.16	177	17	GPL RPC xdrccp info query
RT	9	bob-virtua...	3.3342	2019-09-25 03:25:31	10.10.10.50	37618	10.10.10.16	445	6	GPL NETBIOS SMB.OS IPCS share access
RT	105	bob-virtua...	3.3202	2019-09-19 15:02:01	10.10.10.50		10.10.10.16		1	GPL ICMP_INFO PING 'NAX
RT	81	bob-virtua...	3.3474	2019-12-24 06:21:05	10.10.10.79		10.10.10.16		1	GPL ICMP_INFO PING 'NAX
RT	1	bob-virtua...	3.3319	2019-09-25 03:25:31	10.10.10.50	38772	10.10.10.16	21	6	GPL FTP PORT bounce attempt

IP Resolution Table:

Sid	Net	Hostname	Status	Type	Last
1	bob...	bob-virtual-ma...	ossec		2019-09...
3	bob...	bob-virtual-ma...	snort		2019-12...
2	bob...	bob-virtual-ma...	pcap		2020-01...

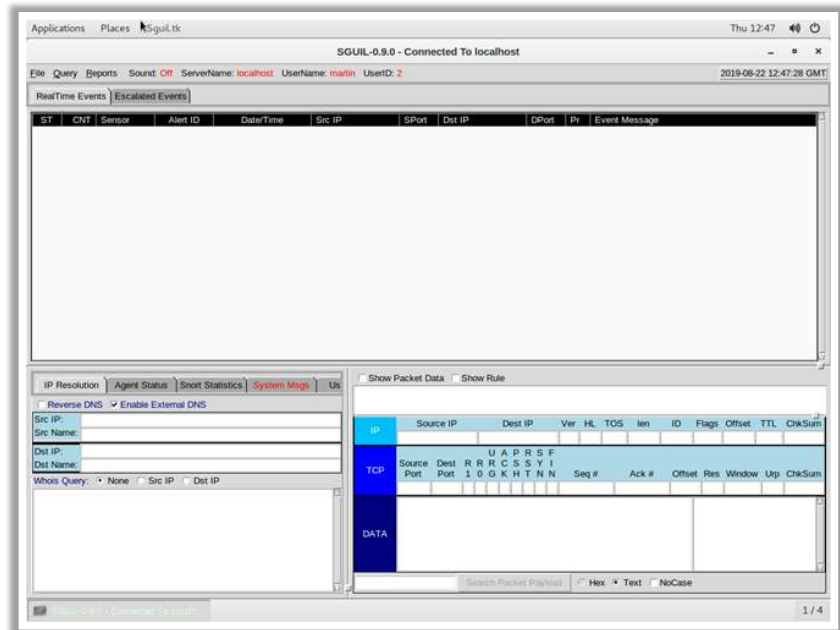
Display Detail:

User: bob
Sep 19 12:42:15 bob-Virtual-Machine sudo: pam_unix(sudo:auth): authentication failure; logname= uid=1000
euid=0 tty=/dev/pts/1 ruser=bob rhost= user=bob
uid=1000
euid=0
tty: /dev/pts/1

Command Console

Sguil Command Console

- ❑ Command console software is installed and runs on a **separate system** that is dedicated to the IDS
- ❑ It provides a **user interface** to an administrator for the purpose of receiving and analyzing security events, alert message, and log files
- ❑ It evaluates **security event** information from different security devices
- ❑ **Caution:** If the command console is installed on a non-dedicated computer system (e.g., firewall, backup server), it will drastically slow down the response to security events as those systems may be busy handling other tasks



Alert Systems



An alert system sends an **alert message** when any anomaly or misuse is detected



OSSEC HIDS Alerts in Sguil

SGUIL-0.9.0 - Connected To localhost

File Query Reports Sound: Off ServerName: localhost Username: martin UserID: 2 2020-01-03 12:39:25 GMT

RealTime Events | Escalated Events

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	3	bob-virtua...	1.1752	2019-09-19 02:14:51	0.0.0.0		10.10.10.16		0	[OSSEC] Windows: System time changed.
RT	30	bob-virtua...	1.1402	2019-09-09 08:23:05	0.0.0.0		10.10.10.16		0	[OSSEC] Windows: Logon Failure - Unknown us...
RT	2	bob-virtua...	1.1749	2019-09-19 01:09:34	0.0.0.0		10.10.10.16		0	[OSSEC] The audit log was cleared
RT	2	bob-virtua...	1.1751	2019-09-19 01:42:16	0.0.0.0		10.10.10.16		0	[OSSEC] PAM: User login failed.
RT	2	bob-virtua...	1.1748	2019-09-19 01:08:48	0.0.0.0		10.10.10.16		0	[OSSEC] Listened ports status (netstat) change...
RT	60	bob-virtua...	1.1653	2019-09-19 01:00:00	0.0.0.0		10.10.10.16		0	[OSSEC] Integrity checksum changed.
RT	4	bob-virtua...	1.1649	2019-09-19 00:58:55	0.0.0.0		10.10.10.16		0	[OSSEC] Host-based anomaly detection event (...)
RT	35	bob-virtua...	1.1657	2019-09-19 01:00:03	0.0.0.0		10.10.10.16		0	[OSSEC] File added to the system.
RT	1	bob-virtua...	3.3695	2019-12-24 12:11:06	10.10.10.50	61121	10.10.10.16	177	17	GPL RPC xdmcp info query
RT	3	bob-virtua...	3.3342	2019-09-25 03:25:31	10.10.10.50	37618	10.10.10.16	445	6	GPL NETBIOS SMB-D5 IPCS share access
RT	105	bob-virtua...	3.3202	2019-09-19 15:02:01	10.10.10.50		10.10.10.16		1	GPL ICMP_INFO PING *NIX
RT	81	bob-virtua...	3.3474	2019-12-24 06:21:05	10.10.10.79		10.10.10.16		1	GPL ICMP_INFO PING *NIX
RT	1	bob-virtua...	3.3319	2019-09-25 03:25:31	10.10.10.50	38772	10.10.10.16	21	6	GPL FTP PORT bounce attempt

IP Resolution Agent Status Short Statistics System Mags Us

Sid	Net	Hostname	Status	Type	Us
1	bob...	bob-virtual-ma...	ossec		2019-09...
3	bob...	bob-virtual-ma...	snort		2019-12...
2	bob...	bob-virtual-ma...	pcap		2020-01...

Update Interval (secs): 15 NOW

Display Detail

User: bob
Sep 19 12:42:15 bob-Virtual-Machine sudo: pam_unix(sudo_auth): authentication failure; logname= uid=1000
euid=0 tty=/dev/pts/1 user=bob host= bob-virtua...
uid=1000
tty: /dev/pts/1

Snort NIDS Alerts in Sguil

SGUIL-0.9.0 - Connected To localhost

File Query Reports Sound: Off ServerName: localhost Username: martin UserID: 2 2019-08-22 13:48:33 GMT

RealTime Events | Escalated Events

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	2	bob-virtua...	3.94	2019-08-22 13:44:29	10.10.10.50	50254	10.10.10.16	3306	6	ET SCAN Suspicious inbound to mySQL port 3306
RT	1	bob-virtua...	3.96	2019-08-22 13:44:29	10.10.10.50	47824	10.10.10.16	5907	6	ET SCAN Potential VNC Scan 5900-5920
RT	2	bob-virtua...	3.97	2019-08-22 13:44:30	10.10.10.50	46220	10.10.10.16	5432	6	ET SCAN Suspicious inbound to PostgreSQL port ...
RT	2	bob-virtua...	3.99	2019-08-22 13:44:30	10.10.10.50	43738	10.10.10.16	1433	6	ET SCAN Suspicious inbound to MSSQL port 1433
RT	1	bob-virtua...	3.101	2019-08-22 13:44:30	10.10.10.50	36150	10.10.10.16	5800	6	ET SCAN Potential VNC Scan 5800-5820
RT	2	bob-virtua...	3.102	2019-08-22 13:44:31	10.10.10.50	43824	10.10.10.16	1521	6	ET SCAN Suspicious inbound to Oracle SQL port 1...

IP Resolution Agent Status Short Statistics System Mags Us

Show Packet Data Show Rule

Reverse DNS Enable External DNS

Src IP: Src Name: Dst IP: Dst Name: Whos Query: None Src IP Dst IP

IP	Source IP	Dest IP	Vw	HL	TOS	len	ID	Flags	Offset	TTL	ChkSum
TCP	Source Port	Dest Port	U	A	P	R	S	F			
			1	0	0	K	H	T	N		
			Seq#	Ack#	Offset	Res	Window	Up	ChkSum		

DATA

Search Packet Payload Hex Text NoCase

Response System



The response system issues **countermeasures** against any intrusion that is detected



You also need to be involved in the decision during incident response and should have the ability to respond on your own. You need to make **decisions** on how to deal with false positives and when a response needs escalation



Recommendation: You should not solely rely on an IDS response system for an intrusion response



Attack Signature Database



An IDS does not have the capability to make a decision, instead it maintains a **database** of attack signatures and patterns



Network traffic is compared against these known **attack signatures** and then a decision can be made

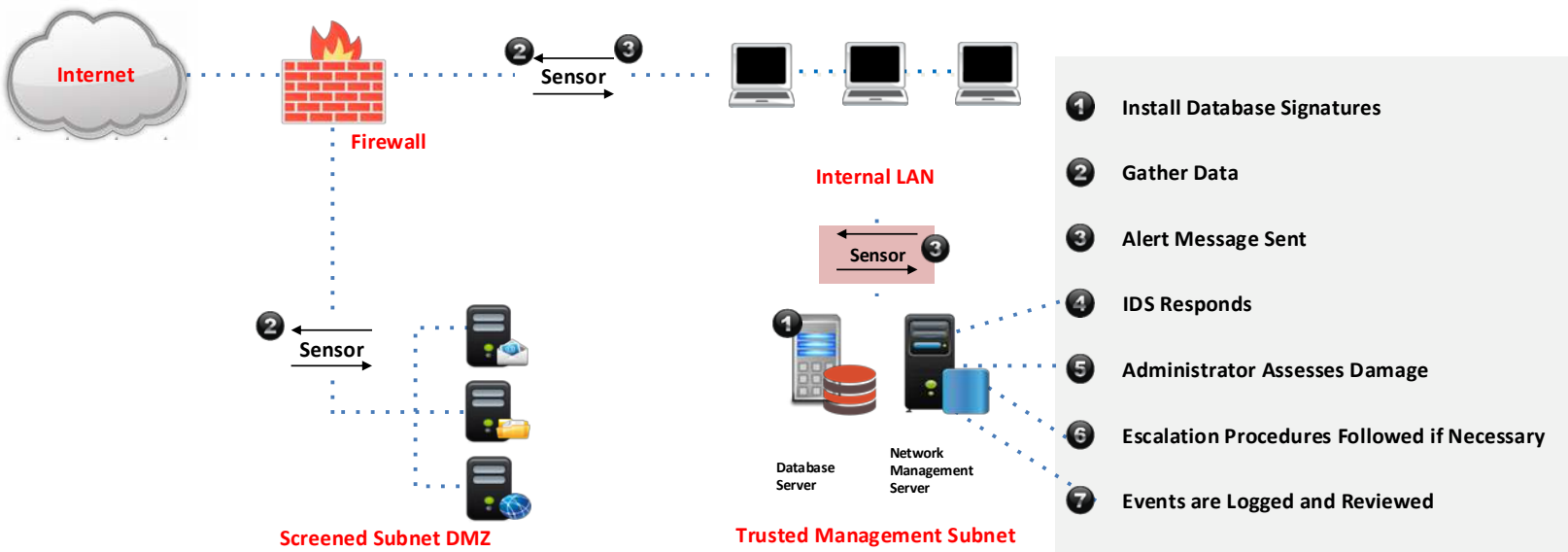


If any matches are found, the IDS will raise an alert and **block** the suspicious traffic



Recommendation: You need to periodically update the IDS attack signature database

Collaboration of IDS Components in Intrusion Detection



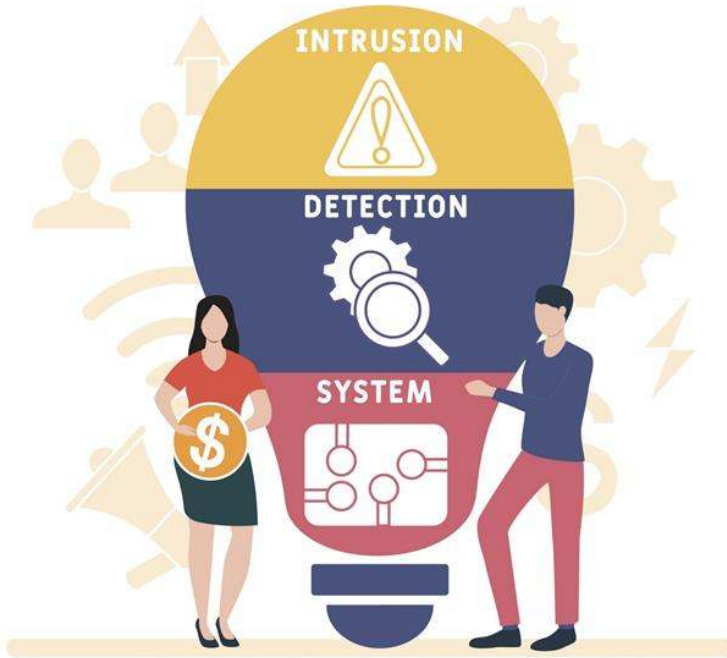
Deployment of Network and Host-based IDS

Staged IDS Deployment

- ❑ You should **plan** for a staged IDS deployment in their network
- ❑ A staged deployment will help you gain **experience** and **discover** how much monitoring and maintenance of network resources is actually required
- ❑ The **monitoring** and **maintenance** of network resources varies depending on the size of an organization's network



Deploying Network-based IDS



An effective deployment of NIDS requires a lot of attention concerning the **network topology** of the organization

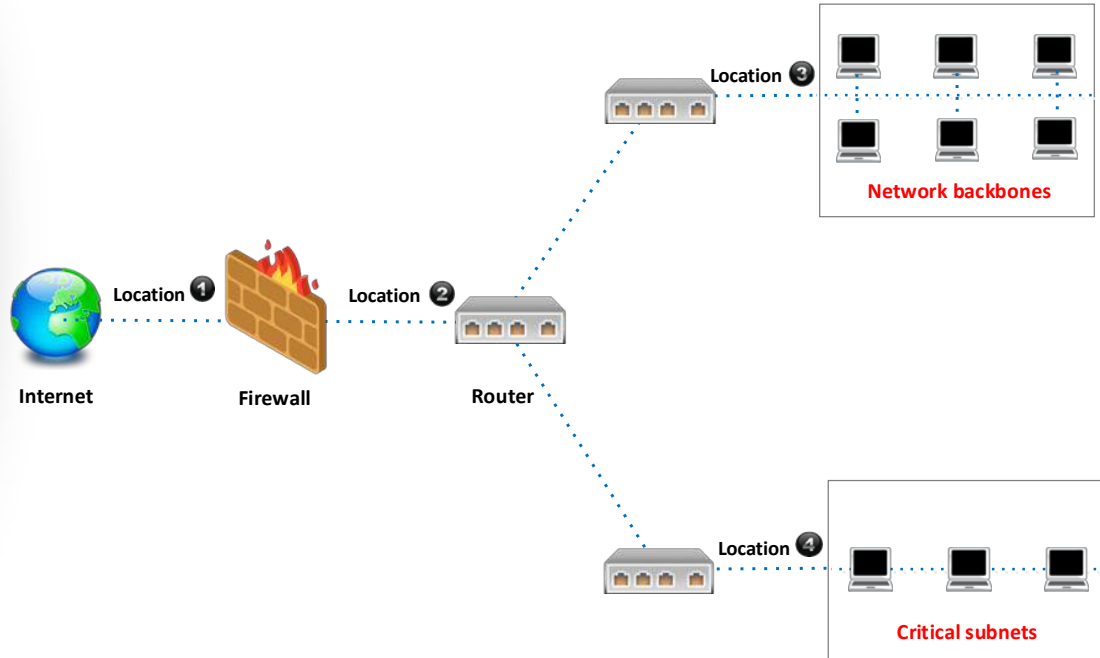


The possible IDS deployment options are categorized based on the location of IDS sensors



Consider all possible **options** and its associated **advantages/disadvantages** when placing a network-based IDS

Deploying Network-based IDS (Cont'd)



Deploying Network-based IDS (Cont'd)



Location 1

- ❑ Place an IDS sensor behind each **external firewall** and in the **network DMZ**

Advantages

- ✓ Monitors attacks originating from the outside world
- ✓ Highlights the inability of the firewall and its policies to defend against attacks
- ✓ It can see attacks which target the web or FTP servers located in the DMZ
- ✓ Monitors outgoing traffic results from a compromised server



Location 2

- ❑ Place an IDS sensor outside an **external firewall**

Advantages

- ✓ Ability to identify the number and types of attack originating from the Internet to the network



Deploying Network-based IDS (Cont'd)



Location 3

- ❑ Place an IDS sensor on major **network backbones**

Advantages

- ✓ Monitors and inspects large amounts of traffic, increasing the chance for attack detection
- ✓ Detects unauthorized attempts from outside the organization



Location 4

- ❑ Place an IDS sensor on **critical subnets**

Advantages

- ✓ Detects attacks on critical systems and resources
- ✓ Focuses on specific critical systems and resources

Deploying a Host-based IDS



Deploying a host-based IDS provides an **additional layer of security**



This type of IDS must be installed and configured on **each critical system** in the network



You should consider installing a host-based IDS on **every host** in the organization



When deploying a host-based IDS, it is recommended that it has **centralized management** and **reporting** functions, which reduces the complexity for managing alerts from a large number of hosts



What is an IDS Alert?

- ❑ Alert is a **graduated event**, which notifies that a particular event (or series of events) has reached a specified threshold and needs proper action by a responsible party



- ❑ It sends the notification, indicating that something is wrong and **requires immediate attention** and monitoring

Types of IDS Alerts



True Positive (Attack - Alert)

- ☐ An IDS raises an alarm when a **legitimate attack** occurs



False Positive (No Attack - Alert)

- ☐ An IDS raises an alarm when **no attack** has taken place



False Negative (Attack - No Alert)

- ☐ An IDS does not raise an alarm when a **legitimate attack** has taken place



True Negative (No Attack - No Alert)

- ☐ An IDS does not raise an alarm when an **attack** has not taken place



Characteristics of Good IDS Solutions

- 01 Run continuously with **less human intervention**
- 02 Must be **fault tolerant**
- 03 Resistant to **subversion**
- 04 **Minimal overhead** on the system
- 05 Observe **deviations** from normal behavior
- 06 Not easily **deceived**
- 07 Tailored to specific **system needs**
- 08 Copes with **dynamic system behavior**



Selection of an Appropriate IDS/IPS Solutions

- ❑ IDS products must meet certain **criteria** to be deployed in an organization
- ❑ Compare the different technology types, then select the most appropriate **technology** to meet the requirements

The products should be evaluated based on organizational requirements such as:

General requirements

- ❑ Evaluate the general requirements the IDS products will have to meet post deployment
- ❑ Size of an organization also modifies the number of IDS products needed

Security Capability Requirements

- ❑ The selection of an IDS depends on an organization's environment and policies as well as the current security and network infrastructure

Performance requirements

- ❑ Evaluate an IDS product's general performance characteristics by assessing its capacity to handle the network traffic or packet monitoring capabilities for NIDS and event monitoring capabilities for HIDS

Management requirements

- ❑ The products need to comply with the organization's management policy in order to be used effectively

Life Cycle Costs

- ❑ Estimated lifecycle costs of the products should be within the available budget

Intrusion Detection with Snort

01

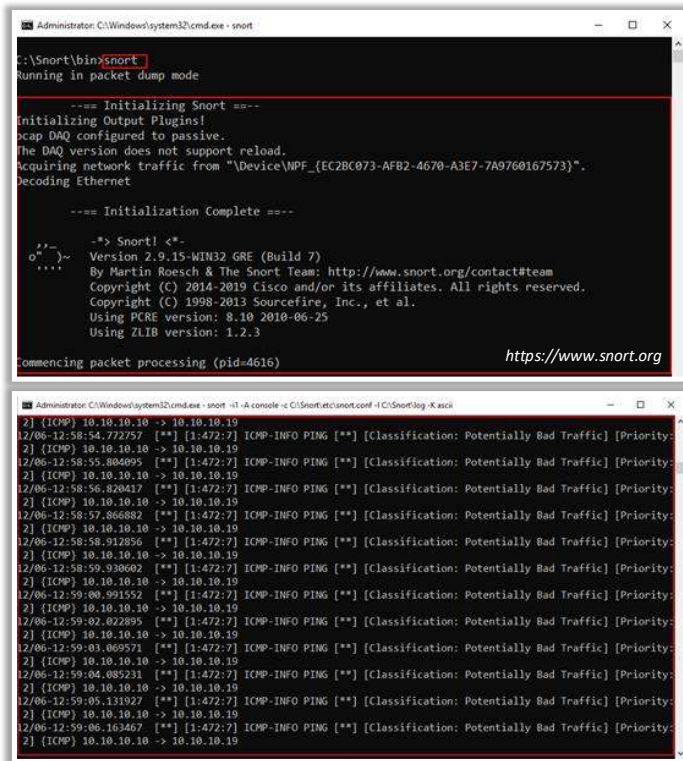
Snort is an open-source network intrusion detection system, capable of performing real-time **traffic analysis and packet logging on IP networks**

02

It can perform **protocol analysis and content searching/matching**, and is used to detect a variety of **attacks and probes**, such as buffer overflows, stealth port scans, and OS fingerprinting attempts

03

It uses a flexible **rules language** to describe traffic that it should collect or pass, as well as a **detection engine** that utilizes a modular plug-in architecture



```
Administrator: C:\Windows\system32\cmd.exe - snort

C:\Snort\bin>snort
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
The DAQ version does not support reload.
Acquiring network traffic from '"Device\NPF_{EC2BC073-AFB2-4670-A3E7-7A9760167573}"'.
Decoding Ethernet

--== Initialization Complete ==--

--* Snort! <*-
o~ Version 2.9.15-WIN32 GRE (Build 7)
**** By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
      Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
      Copyright (C) 1998-2013 Sourcefire, Inc., et al.
      Using PCRE version: 8.10 2010-06-25
      Using ZLIB version: 1.2.3

commencing packet processing (pid=4616)

https://www.snort.org

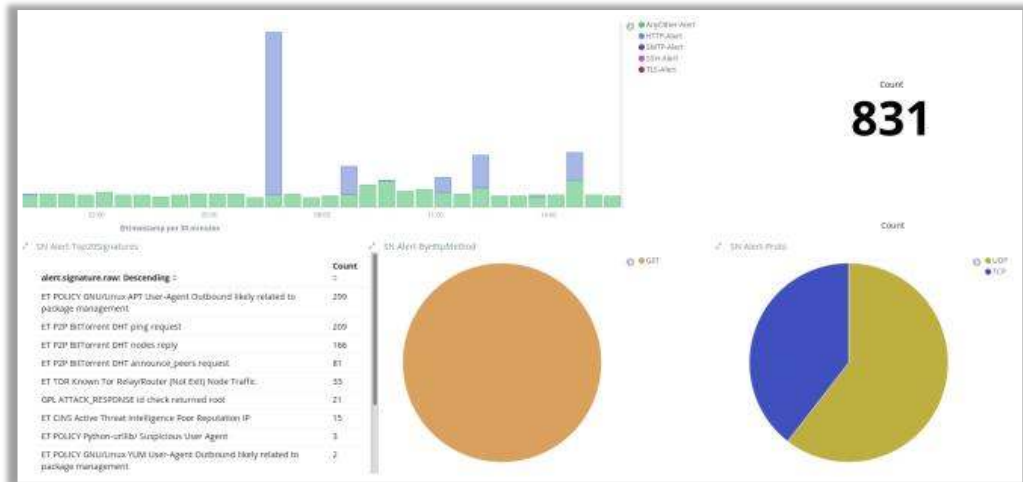
Administrator: C:\Windows\system32\cmd.exe - snort -i 1 -A console -c C:\Snort\etc\snort.conf -l C:\Snort\log -K ascii

2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:58:54.772757 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:58:55.880495 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:58:56.820417 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:58:57.866882 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:58:58.912856 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:58:59.930602 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:59:00.991552 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:59:02.022895 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:59:03.069571 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:59:04.085231 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:59:05.131927 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
12/06-12:59:06.163467 [**] [1:472:7] ICMP-INFO PING [**] [Classification: Potentially Bad Traffic] [Priority:
2) [ICMP] 10.10.10.10 -> 10.10.10.19
```

Intrusion Detection Tools

Suricata

Suricata is a robust network threat detection engine capable of **real-time intrusion detection (IDS)**, **inline intrusion prevention (IPS)**, **network security monitoring (NSM)**, and **offline pcap processing**



<https://suricata-ids.org>



AlienVault® OSSIM™

<https://cybersecurity.att.com>



SolarWinds Security Event Manager

<https://www.solarwinds.com>



OSSEC

<https://www.ossec.net>



Zeek

<https://zeek.org>



Sagan Log Analysis Engine

<https://quadrantsec.com>